Combining explicit Diophantine encodings of ZFC consistency

Research report prepared for Vladimir Reshetnikov

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Abstract

Eight independent packages encode the same blank K215 computation, but optimize different parts of the translation. We combine the inputless construction of packages 06/08 with the ordinary strong Pell equation of package 02. The result is six equations in fifteen positive unknowns, of maximum degree 28, evaluated in 61 binary arithmetic operations. A single product polynomial has degree 99 and costs 69 operations; a sum of squares has degree 56 and costs 78. Further independent improvements give a quartic in 33 positive unknowns, an equation in 31 unrestricted integer unknowns, and a smaller finite local language for the fixed coefficients. These are mathematical constructions with reproducible algebraic and finite checks, not newly kernel-checked composite Lean or Rocq proofs. No global minimum is asserted.

1 What is being minimized?

The requested equivalence is an internal theorem:

$$\mathrm{ZFC} \vdash \big(\text{the displayed equation has no solutions in its stated domain}\big) \longleftrightarrow \mathrm{Con(ZFC)}.$$

Equivalently, existence of a solution is equivalent to existence of a finite ZFC refutation. Neither statement asserts consistency. We keep the literal K215 machine and its existing calibration, rather than changing the foundational theory.

There is no single ordering of "simplest": variable count, degree, equation count, arithmetic-circuit size, coefficient height, and proof complexity conflict. We therefore give explicit alternatives, not a claim that one is smallest in every sense. An operation means a binary addition, subtraction, or multiplication, even by a fixed numeral. Fixed coefficients, reuse, and comparisons are free. Coefficient bit length is *not* included in that cost.

Construction	Unknowns/domain	Equations	Degree	Operations
Previous own construction	18 positive	1	187	82
Existing 06/08 product	15 positive	1	155	67
New ordinary system	15 positive	6	28 maximum	61
New ordinary product	15 positive	1	99	69
New ordinary SOS	15 positive	1	56	78
New quadratic system	33 positive	24	2	not optimized
New quartic SOS	33 positive	1	4	not optimized
New signed product	31 integer	1	188	127
New signed SOS	31 integer	1	100	136

The 67-operation product remains preferable if arithmetic count alone is the objective. The new 61-operation system improves degree without increasing its count; the 69-operation product pays two operations for a substantial degree reduction. All fixed numerals are specified by finite recipes, not by waiting for K215 to halt.

2 Comparison of the eight inputs

The reviewed containing commit is `3b201cdea32d2c816f5f3d20734051f25fd3ef14`. The shared original arithmetic source is pinned at `cdd582bec13f908df3711fa9e05d1ddac368a5db`. Package numbers below refer to the repository's reconciliation directory, not to a chronological priority claim.

Packet	Useful contribution	What to retain or improve
01	Conservative 82-operation specialization; separate Lean tag-system route.	Good independent dependency route, but retaining the numerical input bridge costs witnesses and gates.
02	Ordinary rather than normalized strong norm; 16-variable, seven-equation, degree-28 system.	Import the large-index recovery proof, not an unjustified same-tuple algebraic substitution.
03	Alternative 16-variable degree-28 system and 15-variable substitutions.	Eliminating a variable by a large polynomial substitution can sharply increase degree.
04	Two-sided normalizer and factored product.	Useful interface comparison; raw blank K215 avoids its normalization overhead.
05	Several 15/16-variable variants and proof tradeoffs.	Keep their metrics distinct: a smaller witness list need not mean a smaller circuit or degree.
06	Raw K215, inputless language, exact coefficient recipe; 15-variable product 67.	Best simple starting backend; its weaker rank bound includes the preliminary $C = 0$ case.
07	Short 70-operation fixed-input product and retained-index system.	Retaining index witnesses can help proof structure but is not needed in our final core.
08	Same inputless arithmetic as 06, plus a 34-variable quadratic lift.	Import the lift methodology, then remove one auxiliary and one equation.

The accompanying comparison artifact records the individual headline metrics and their provenance. Intake into the repository did not run the delivered verifiers or formal proof builds. We do not treat an intake claim as a new verification result. In particular, similarly named mask coefficients are not interchangeable: the ordering of Start, a forbidden window, payload copies and anchors is part of the theorem.

3 The fifteen-variable system and two single equations

Supply exactly

$$(J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma) \in \mathbb{Z}_{>0}^{15}.$$

The four *fixed* coefficients are $\beta = B - 1$, K, M_C, and $M_F^{\mathrm{src}} = M_F^{\mathrm{nat}} + B - 1$. They are not extra unknowns. Define polynomial abbreviations

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H, & D &= X + ac + \gamma H. \end{aligned}$$

Then put

$$\begin{aligned} G &= (q-1)(q-F) + \alpha, \\ R &= G(q^2-1) + (M_C + qM_F^{\mathrm{src}})J, \\ L_0 &= EkY, \qquad V = c(Tf - 1) - Rf^2, \\ I &= ic^2, \qquad A_0 = \Delta(f^2 - 1). \end{aligned}$$

Every exponent here is fixed. No variable exponent, division, bit operation, or computation predicate occurs in the polynomial. Define

$$N_0 = \tau^2 - L_0(L_0 + k), \qquad N_1 = D^2 - \Delta c^2,$$
$$N_A = A_0(V^2 - y^2) + y^2, \qquad N_K = k - hE - R,$$
$$N_T = (K + w)C + q - F - z(q - 1), \qquad N_S = 1 + I^2 - A_0.$$

The recommended small system is

$$\boxed{N_0 = N_1 = N_A = N_K = N_T = 1, \qquad I^2 = A_0.} \tag{1}$$

Two explicit single polynomials are

$$P_{69} = N_0 N_1 N_A N_K N_T N_S - 1, \tag{2}$$
$$S_{78} = (N_0 - 1)^2 + (N_1 - 1)^2 + (N_A - 1)^2 + (N_K - 1)^2 + (N_T - 1)^2 + (I^2 - A_0)^2. \tag{3}$$

The subscript is the complete binary-operation count, not a state count or degree.

Theorem 1. *For the fixed coefficient recipe specified below, existence of a positive solution of any of (1), $P_{69} = 0$, or $S_{78} = 0$ is equivalent to blank K215 halting, hence to a finite ZFC refutation. The equivalences are formalizable in ZFC.*

3.1 Why the ordinary strong equation is a real improvement

The normalized core uses $(i\Delta c^2)^2 = \Delta(f^2 - 1)$ and a high-degree auxiliary coefficient. We use $(ic^2)^2 = \Delta(f^2 - 1)$. A normalized solution gives an ordinary one by replacing i with Δi. The converse is *not* justified by dividing i by Δ: its integrality is not assumed. Instead a large-index Pell theorem recovers the needed rank and divisibility directly. This is the substantive combination of 02 with 06/08.

3.2 Soundness, with no circular positivity assumption

At a zero of (2), every factor is an integer unit. The negative Pell equation for N_0 is impossible by descent. Since $\Delta \equiv 0, 3 \pmod 4$, N_1 cannot be -1; also $A_0 \equiv 0, 1 \pmod 4$, excluding -1 for N_S and N_A. Thus these four factors are 1, while $N_K = \varepsilon$, $N_T = \nu$ with $\varepsilon\nu = 1$.

Transport implies only $C \geq 0$ at first, not $C > 0$. Using the authentic mask bounds yields

$$q(q^2 - 1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \qquad E \geq q^4 > R + 2.$$

The weaker upper bound, including the $C = 0$ boundary, is deliberate. Let $A = a + 2$, and define the Pell recurrences by

$$\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n.$$

The first and main norm equations give, with $P_0 = 2XY^2 + 1$,

$$k = 2\psi_{P_0}(n), \quad 2n \equiv R + \varepsilon \pmod E, \quad c = \psi_A(p), \quad p > n \geq (R - 1)/2.$$

Their growth proves $c > A\Delta^2$, $c > 2p$ and $c > 2R$, before using the auxiliary equation to identify p.

The ordinary strong-rank lemma now gives

$$f = \chi_A(m), \quad I = \Delta\psi_A(m), \quad p \mid m, \quad c \mid m, \quad c^2 \mid f^2 - 1, \qquad m \geq c > 2p.$$

This lemma handles the apparent rational Pell coefficient I/Δ using squarefree reduction and strong divisibility; the proof and precise imported source are included in the algebra proof artifact. It cannot be replaced by informal division of the equation.

Next $f > 2c$, and $A_0 > Rf^2 + c$. The auxiliary norm forces $V > 1$; the congruence $V \equiv -c \pmod f$ excludes $0, \pm 1$, and its Pell gap excludes a negative V. The omitted positive quotients are recovered as

$$o = cT - Rf, \qquad j = Tf - 1 - R(f^2 - 1)/c, \qquad V = of - c = jc - R.$$

Their integrality uses the just-proved divisibility. Signed auxiliary step-down, with $0 < p, R < c/2$, gives $p = R$. The first-norm index congruence cannot wrap because $E > R + 2$, and the strict ratio $kY < c < k(Y + 1)$ excludes $\varepsilon = -1$. Hence $\varepsilon = \nu = 1$, and transport now excludes $C = 0$.

The lower Pell-ratio and projected recurrence give $X = 2^R$. The half-binomial population identity, repunit congruence, inverse-population packing and native synchronization recover a marked periodic computation tableau. A rectangle through Start gives a genuine blank halting K215 run. These are the existing compiler lemmas with the explicit local-language adaptation below, not consequences of finite numerical tests.

3.3 Completeness and exact degree

A finite halting computation supplies a padded periodic tableau. The native compiler tunes the actual packed index using dummy bits, then chooses its Pell witnesses. This gives the normalized core; the map $i_{\text{ordinary}} = \Delta i_{\text{normalized}}$ gives (1). It proves existence equivalence, not equality of the two sets of tuples.

For exact degrees put $Q = \beta J$, $k_0 = \eta + \zeta$, $C_1 = Q - F - \alpha$. The factor degrees are

$$22, 18, 28, 7, 2, 22.$$

The apparently larger leading terms of the main norm cancel; explicitly

$$D^2 - \Delta c^2 = 2ac(X + \gamma H) + (X + \gamma H)^2 - Hc^2.$$

The degree-99 part of P_{69} is

$$8Q^{55} h\gamma i^2 k_0^9 w^7 s^{16} T^2 f^4 (wC_1 - zQ).$$

Its coefficient at $J^{56} h\gamma i^2 \eta^9 w^7 s^{16} T^2 f^4 z$ is $-8\beta^{56} \neq 0$. Only the auxiliary residual has degree 28, proving $\deg S_{78} = 56$. The JSON circuits and independent checks record every gate.

4 Reducing the unrestricted integer domain to 31 unknowns

This is a separate domain improvement, not a claim that the 33-variable quartic below already works over arbitrary integers. In the fifteen-port ordinary core, only

$$J, F, \alpha, z, i, s, w, \eta, \zeta$$

need remain positive. The other six ports h, f, T, γ, τ, y may independently be signed.

The unit extraction and raw rank bounds do not use those six signs. First $h < 0$ would give $k \leq R + 1 - E < 0$. If $h = 0$, then $k = R \pm 1 > 2$, while the first Pell sequence has first value $k = 2$ and second value $k = 4P_0 > R + 1$. Hence $h > 0$ automatically.

Classify the main norm by $|D| = \chi_A(p)$, and the strong norm by $|f| = \chi_A(m)$. The earlier bootstrap and large-rank lemma still apply. In the auxiliary norm classify $I|V|$, rather than assuming $V > 0$. Its odd-index quotient polynomial gives the signed congruences needed for $R \equiv \pm p \pmod c$. Since

$0 < R, p < c/2$, these still imply $R = p$. Thus no positive quotient reconstruction is required for signed f, T.

The sign of D can also be recovered, rather than imposed. Put $r = (R-1)/2$. Standard Pell growth, independently of the sign of D, gives

$$\frac{2c}{k} \geq \left(\frac{(2A-1)^2}{2P_0}\right)^r > X^r, \qquad \frac{2c}{k} < 2Y + 2, \qquad Y > X^r/3.$$

If $D < 0$, reducing the Pell identity modulo $H = 4a + 3$ would yield

$$H \mid X2^R + 1.$$

But $X \geq 16$ and the ratio bound give $0 < X2^R + 1 < H$, a contradiction. Thus $D > 0$, $X = 2^R$, and

$$\gamma H = \chi_A(R) - ac - X > c - X > 0.$$

The downstream arithmetic and tableau decoding now applies unchanged. Completeness follows from the positive construction. This is an existence equivalence, not a coordinatewise absolute-value map.

For each of the nine positive ports use the explicit parametrization

$$x = 1 + u_x^2 + v_x^2 + r_x(r_x + 1), \qquad u_x, v_x, r_x \in \mathbb{Z}.$$

It is always positive and represents every positive integer: apply Legendre's three-square theorem to $4(x-1)+1$, whose three roots have exactly one odd member. Keep the six signed ports unchanged. The resulting $9 \cdot 3 + 6 = 33$ unrestricted integer variables give the claimed system and single equations without hidden side conditions.

Each substitution costs seven operations. Assigning weight two to the nine replaced ports and weight one to the six signed ports gives factor weights

$$44, 35, 50, 13, 4, 44.$$

Hence the signed product has exact degree 190 and 132 operations; the SOS has exact degree 100 and 141 operations; the six-equation system has maximum degree 50 and 124 operations. The signed proof artifact contains the full odd-index step-down argument and uniform leading forms.

4.1 Two further coordinates saved by choosing an odd auxiliary

The nine-positive-port construction above is a useful intermediate result, but i need only be nonzero for soundness: it occurs only squared. Replace it by

$$i = 2j_0 + 1, \qquad j_0 \in \mathbb{Z}.$$

This specialization requires a fresh completeness proof, not the assertion that every positive i is odd.

The canonical compiler witness has $R \equiv 3 \pmod 4$, even A, odd Δ, and odd $c = \psi_A(R)$. Choose the strong index $m = cR$, rather than $2cR$, and set

$$f = \chi_A(cR), \qquad t = \frac{\psi_A(cR)}{c^2}, \qquad i = \Delta t.$$

The quotient t is a positive integer: expanding $(D + c\sqrt{\Delta})^c$ shows that its square-root coefficient is divisible by c^2. Since cR is odd and A even, the recurrence makes $\psi_A(cR)$ odd. Thus t and i are odd.

Write $R = 2r + 1$ with r odd, and use the integer polynomial Q_r defined by $\chi_Z(2r+1) = ZQ_r(Z^2)$. Set

$$I = ic^2, \qquad V = Q_r(I^2), \qquad y = \psi_I(R).$$

The identities $Q_r(1-A^2) = -\psi_A(R) = -c$ and $Q_r(0) = -R$ give $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$. Therefore $o = (V+c)/f$ is positive integral, and $f^2 \equiv 1 \pmod c$ proves $T = (o+Rf)/c$ is positive integral. The exact compressed auxiliary formula is restored. No parity restriction on the old auxiliary index was silently carried over.

Only eight ports now use the three-coordinate positive parametrization; j_0 and the other six signed ports are free. This gives $8 \cdot 3 + 1 + 6 = 31$ unrestricted integers. The added cost is $8 \cdot 7 + 2 = 58$ operations. Factor weights become $44, 35, 50, 13, 4, 42$, giving product degree 188 and SOS degree 100. The complete costs are 119 for the six equations, 127 for the product and 136 for the SOS. The signed31 JSON files spell out all substitutions and gates.

5 A quartic in 33 positive unknowns

We give two different 33-variable quadratic lifts in the archive. The following normalized-strong version is useful because its new algebraic step needs no relaxed strong-rank lemma. It improves the delivered 34-variable, 25-equation lift to 33 variables and 24 equations.

All of the following coordinates are strictly positive:

$$J, F, \alpha, z, f, h, i, o, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, A_1, G, L, R, V, g, I, B_1, U, H_1, j.$$

Only $q = \beta J + 1$ and $k = \eta + \zeta$ are abbreviations. The complete system is

$$\begin{aligned}
L &= q^2 - 1, & N &= (L+1)q,\\
X &= wq, & a &= E + Ns,\\
E &= X(a-E), & c &= \eta + (a-E)k,\\
P &= E(c-\eta) + k, & \tau^2 &= P(P-k) + 1,\\
D &= X + ac + \gamma(4a+3), & A_1 &= D^2 - 1,\\
U &= (a+1)c, & A_1 &= U(U+2c),\\
k &= Eh + R + 1, & G &= (q-1)(q-F) + \alpha,\\
R &= GL + J(M_C + qM_F^{\mathrm{src}}), & &\\
(K+w)(q-F-\alpha) + q - F - z(q-1) &= 1, & &\\
H_1 &= (a+1)(f-1), & V &= of - c,\\
V &= jc - R, & g &= y^2 - V^2,\\
I &= A_1 i, & B_1 &= I^2,\\
B_1 &= H_1(H_1 + 2a + 2f + 4), & y^2 - B_1 g &= 1.
\end{aligned}$$

After replacing the two linear abbreviations, every residual has degree at most two. The sum of their 24 squares is a single polynomial Q_{33} of exact degree four: the coefficient of τ^4 is exactly one. These equations, not an unspecified circuit-lifting algorithm, define the quartic completely.

The saving comes from two factorizations:

$$U(U+2c) = (a+1)(a+3)c^2 = \Delta c^2,$$
$$H_1(H_1 + 2a + 2f + 4) = (a+1)(a+3)(f^2-1) = \Delta(f^2-1).$$

The old three-variable norm block becomes a two-variable block. Replacing the merged auxiliary quotient by o, j balances the other additions/removals.

For projection to the normalized core, $B_1 = I^2 = \Delta^2 c^4 i^2 = \Delta(f^2 - 1)$ gives $f^2 \equiv 1 \pmod{c}$. From $of + R = (j+1)c$, the merged quotient

$$T = (o + Rf)/c$$

is a positive integer, and $c(Tf - 1) - Rf^2 = V$. Conversely a positive normalized-core solution supplies

$$o = cT - Rf, \qquad j = Tf - R\Delta i^2 c^3 - 1,$$

which are integers and positive because $of = V + c > 0$, $jc = V + R > 0$. The remaining introduced coordinates are explicitly positive. Thus both directions are exact; no new machine run is needed.

The 33-positive-variable quartic is not a 33-variable unrestricted-integer quartic. The repository reports an 83-integer-variable quartic in packet 04; our safe 99-integer degree-eight conversion does not improve that benchmark.

The ordinary-strong lift in the archive instead deletes the old redundant main-norm auxiliary and sets $I = ic^2$. It also has 33 positive coordinates and 24 quadratics, but uses the ordinary-rank proof. All expanded residuals and the two lifting proofs are included. Shifting every positive coordinate by one gives 33 natural variables without changing degree; the three-square parametrization gives a safe 99-integer-variable degree-eight form, distinct from the earlier 33-integer high-degree result.

6 A smaller coefficient backend: four cells, not nine

The arithmetic reductions are independent of the following coefficient improvement. We replace the old nine-cell windows by four-cell patterns and keep raw K215, without a one-sided fold or input-erasing machine. This changes all four fixed coefficients together; it is not a substitution of one historical constant into another packet's circuit.

6.1 Literal finite language

Use 439 cell codes: headless bits $0, 1$; a bit b with head state q has code $2(q+1) + b$ for $0 \le q \le 215$; and initialization/boundary symbols

$$L = 434, \quad Q = 435, \quad R_b = 436, \quad V_b = 437, \quad H_b = 438.$$

Initialization payloads are blank, with a state-zero head only at Q. The literal 430 K215 transitions are retained. Halt-state transitions are completed by staying in place without writing, solely for padding after the real halt. In particular the actual first action is $(0, 0) \mapsto (1, +1, 93)$.

A tile is (l, c, r, n): left, center, right, and next-time center. The executable finite predicate in the archive has these requirements:

1. At most one of l, c, r has a head. The vertical-boundary bit agrees between c, n; adjacent vertical boundaries and adjacent temporal horizontal boundaries are forbidden.

2. Nonvertical c, r agree on being horizontal boundaries. A horizontal-boundary center has an initialization symbol as successor.

3. Initialization phases obey LL, LQ, QR_b, R_bR_b, with L next to a left boundary and R_b next to a right boundary.

4. A head cannot escape through a vertical boundary. Immediately before a horizontal boundary it must be halted.

5. At an interior center, if the next cell is not a horizontal boundary, it is an ordinary, noninitialization cell given by the exact deterministic three-neighbor update. Vertical boundaries supply blank/headless payloads; a horizontal boundary cannot supply an interior update.

Outgoing phase recognition is globally equivalent to "an initialization cell is immediately after a horizontal boundary." It removes the old fifth, previous-time cell. The one-head restriction preserves completeness: pad every finite computation with at least two head-free cells beside each vertical boundary, including across a helical seam.

The ordered alphabet begins with

$$\text{Start} = (434, 435, 436, 1), \qquad \text{Forbidden} = (438, 0, 0, 438),$$

then all other valid tuples in lexicographic order. Forbidden is locally valid but has no global extension: its left horizontal boundary cannot have an ordinary nonvertical right neighbor. Start centers on the initialized state-zero head.

There are exactly 37075 tiles. The complete list is included, not merely defined existentially. Its SHA-256 is

`9c2e473b5dbc5b3dc1468a31aefdbec6a901d6ba1e6b1ec9212698d41e1d84b1`.

Direct enumeration checks 20624220 candidates surviving the horizontal head filter; a separate class count agrees.

Only three overlap groups are required. If the right neighbor has index $i-1$ and temporal neighbor index $i-h_0$, impose

$$w_i(r) = w_{i-1}(c), \quad w_i(c) = w_{i-1}(l), \quad w_i(n) = w_{i-h_0}(c).$$

Writing $a_i = w_i(c)$, the tile is exactly $(a_{i+1}, a_i, a_{i-1}, a_{i-h_0})$. Thus the cyclic word pulls back to the plane $a_{-x-h_0 t}$ for *any* stride, even with residue collisions. No coprimality or divisibility assumption is hidden.

A marked periodic plane gives a finite computation strip: the phase graph cannot cycle through Q, so there are vertical boundaries on both sides; periodicity gives a subsequent horizontal boundary; deterministic updates then give a blank run ending in halt. Other Starts are harmless. Conversely any finite halting run admits independent width and height padding and a helical cyclic presentation with one canonical Start.

6.2 The four exact numerals

Here is a closed finite recipe, with no free machine index. Put

$$A_c = 65536, \quad \rho = 2^{390625}, \quad B = \rho^{9765625} = 2^{5^{18}}.$$

For $0 \le u \le 438$, define

$$e_l(u) = 40587 + u, \quad e_c(u) = 40148 + u, \quad e_r(u) = 39709 + u, \quad e_n(u) = 38831 + u.$$

Let $\mathcal{E}$ contain selectors $0, \dots, 37074$, the 1756 copy positions, dummy position 42343, and anchors $43661, 130983, 392949, 1178847$. Thus $|\mathcal{E}| = 38836$. Set

$$\mu = A_c - 2 + \sum_{j=1}^{18743} A_c^j.$$

For tile i with entries (l_i, c_i, r_i, n_i), define

$$\begin{aligned}
v_i &= 1 + A_c^{1+l_i} + A_c^{440+c_i} + A_c^{879+r_i} + A_c^{1318+n_i} + \sum_{t=1}^{4} A_c^{1756+t}, \\
v_{e_l(u)} &= A_c^{1+u}, \quad v_{e_c(u)} = A_c^{440+u}, \\
v_{e_r(u)} &= A_c^{879+u}, \quad v_{e_n(u)} = A_c^{1318+u}, \\
v_{3^{t-1}43661} &= A_c^{1756+t} \ (1 \le t \le 4), \qquad v_{42343} = 0.
\end{aligned}$$

All sums and exponents in this definition are fixed finite integers. Now put

$$\begin{aligned}
D_C &= \rho^{1317} + \rho^{2228468} + \rho^{349288} + \rho^{1047864} \\
&\quad + \sum_{e \in \mathcal{E}} v_e (\rho^{4586163-e} + \rho^{6943861-e}), \\
D_R &= \rho^{2228029}, \\
M_F^0 &= \mu(\rho^{4586163} + \rho^{6943861}) + \sum_{u=0}^{438} \rho^{40148+u} \\
&\quad + \sum_{u=0}^{438} \big(\rho^{2228029+40587+u} + \rho^{2228029+40148+u}\big) + \rho^{392949} + \rho^{1178847}, \\
M_C &= B - 1 - \sum_{e \in \mathcal{E},\ e \neq 1} \rho^e - 2\rho^{42343}, \qquad M_F^{\mathrm{nat}} = M_F^0 + 4.
\end{aligned}$$

The four ports in every main construction are

$$\boxed{\beta = B - 1, \quad K = D_C + BD_R, \quad M_C, \quad M_F^{\mathrm{src}} = M_F^{\mathrm{nat}} + B - 1.}$$

The shifted high mask is essential. These formulas specify actual integers, although their expansions are not emitted.

6.3 Why the new language fits the arithmetic compiler

The native support has 38836 positions and the mask population is balanced:

$$\mathrm{pc}(M_C) = d - 38836, \qquad \mathrm{pc}(M_F^{\mathrm{nat}}) = 38836, \qquad d = 5^{18}.$$

Also $0 < M_C, M_F^{\mathrm{nat}} < B - 1$, $M_C \equiv 2 \pmod 4$ and $M_F^{\mathrm{nat}} \equiv 4 \pmod 8$. Clause padding gives $\mathrm{pc}(\mu) = 18758$; the three overlap groups and two anchor tests account for the remaining mask bits.

The coefficient-mass bound requires only a radix exponent of 299907, while the chosen exponent is 390625. Thus the existing no-carry estimates, including arbitrary bit rotations and the upper dummy bit, remain valid. Two well-separated clause bands guarantee an uncontaminated band. Two anchor pairs force bit alignment and then whole-cell alignment. All 1319 copy/anchor targets and the support rotation conditions are checked explicitly for the actual finite support, and their general argument is given in the bridge proof.

The clause-band gap is $2 \cdot 1178847 + 4 \equiv 2 \pmod 4$. Paired clause contributions therefore cancel modulo five, giving

$$2D_C - D_R \equiv 3 \pmod 5.$$

Consequently the old dummy-bit subset adjustment still tunes the *actual* packed index to the required temporal shift. It is not an unjustified choice of an independent exponent. The canonical origin and positive-slack arguments survive; in particular the canonical rank remains 3 (mod 4), needed in the odd-i construction.

Coefficient language	Number of tiles	Base B
Old nine-cell raw K215	1420903358357108792377	$2^{5^{67}}$
Five-cell cross	38256607897	$2^{5^{36}}$
Four-cell T, head-filtered	37075	$2^{5^{18}}$

Even the new coefficients are not small in ordinary storage terms. The numeral β has exactly 3 814 697 265 625 bits, about 0.477 decimal terabytes; K has 4 685 021 093 751 bits, about 0.586 terabytes. We materialized the tile alphabet and scalar layout, not those numerals. The full adapter proves that these new coefficients satisfy the same blank-input compiler contract used above. The old authenticated 06 recipe is retained as a separate fallback, not mixed into the new tuple.

7 Evidence, reproducibility, and remaining work

The ZIP contains this PDF and editable TeX, all explicit circuits and quadratic residuals, the literal tile alphabet and machine transitions, coefficient recipes, the full combination/domain/bridge proofs, source pins, and reproducible checks. The checks distinguish:

1. Literal circuit identities, operation counts, liveness, symbolic degrees and exact leading forms.
2. Finite local-language enumeration, geometry, mask congruences and halting-control seams.
3. General mathematical Pell, population and tableau soundness/completeness arguments.
4. The existing K215 logical endpoint and its ZFC-formalizable calibration.

Only the first two are replayed by the packaged scripts. Neither a finite audit nor inspection of an endpoint theorem is a new end-to-end kernel proof of the combined construction. No new Lean/Rocq composite proof or internally encoded ZFC derivation is supplied. The repository's existing compiler and logical proofs remain substantive dependencies. The appendix proof documents state their interfaces and exact source locations.

Independent checks compare the new factored equations with the literal circuits on 256 exact integer/rational assignments and reject a deliberate sign mutation. A separate symbolic checker verifies all six residual identities with coefficient ports symbolic. These are tests of the displayed algebra, not hypothetical solutions encoding a ZFC contradiction. The full 37,075-tile enumeration, separate class count, actual-support geometry tests and toy helical seams validate the finite adapter data. See the packaged receipts for the exact scope of each check.

We do not claim a 14-positive-variable construction, fewer than 67 arithmetic operations for the product, or a global minimum. Removing the repunit or ratio slack without its divisibility or positivity condition would not be valid. A useful next formalization target is the small algebraic quadratic-lift equivalence, followed by the finite T4 adapter; those are cleaner bounded interfaces than re-formalizing all number theory at once.

Under consistency, the equations have no standard solutions and ZFC cannot prove their unsolvability by its second incompleteness theorem. Bare consistency alone does not justify claiming that ZFC also cannot prove existence; that stronger assertion needs an appropriate soundness hypothesis.

Primary sources

1. Pinned eight-package collection and reconciliation.
2. Packet 06: raw blank machine, inputless arithmetic, coefficient and completeness proof.
3. Ordinary strong Pell-rank theorem.
4. Packet 08: inputless system and quadratic lifting.
5. Literal K215 endpoint, and ordinary mathematical calibration.